University of Victoria

Department of Electrical and Computer Engineering

SEENG 426 - B01

Lab Project 4

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Aidan Richards - V00990088

MacKenzie Larsen - V00943994

Manraj Minhas - V00998047

Grygorii Karachevtsev - V00983328

Introduction

The goal of this phase was to run a distributed load test on the Vega web app to see how it performs under stress. By simulating up to 200 users at the same time, we wanted to track response times, system throughput, and hardware (ex. CPU and memory). This helps us spot performance bottlenecks and figure out if the app is ready for a real-world production environment.

Preparation

To get realistic results, we set up a Master-Slave testing network using Apache JMeter.

- **Lab Setup:** We used three lab computers on the same network to prevent internet lag. The Master machine hosted the Vega app, and the two Slave machines helped generate the heavy user traffic.
- **Database Setup:** We cloned the repository used to initialize the users and set up the SQL database needed for the user population. Afterwards, we updated roles and ensured the users were being initialized in the JMeter configuration.
- **Monitoring:** The app was connected to the UVic MySQL database. We installed the PerfMon Server Agent on the Master machine to track exactly how much CPU and Memory the server was using during the test.

Load Test

Execution Plan

We ran the test for a full 60 minutes in non-GUI mode to keep the results accurate. The test slowly added users until 200 people were using the app at once. Every user had to log in and visit the Home Page. From there, 50% of the users downloaded the Defense App, and the other 50% downloaded the 50MB resource file.

Test Results % Analysis

- **System Availability:** The system achieved a 100% availability reading. Even with 200 concurrent users pushing 33,220 total requests, the server never crashed and threw exactly 0 errors.
- **System Throughout:** The app handled an average of 9.18 requests per second overall. However, at absolute peak load, the system hit a max bottleneck throughput of 141 requests per second.
- **Performance Drops:** The app struggled heavily with file downloads. Simply loading the Home Page took only 5 milliseconds, but downloading the Defense App took an average of 18.5 seconds (and maxed out at nearly 34 seconds). We also noticed that throughput actually dropped when moving from 150 users (45.67 req/sec) to 200 users (36.79

req/sec), meaning the server was getting overwhelmed and forcing requests to wait in a queue.

Metrics Collected & Their Importance

To accurately evaluate system readiness, we collected the following key metrics:

- **Response time (Avg, Min, Max):** Measures how long the server takes to process specific requests. This is critical for evaluating the end-user experience.
- **Throughput (req/sec):** Measures the volume of requests the server can successfully handle simultaneously, indicating the app's overall capacity.
- **System Resource Utilization (CPU & Memory):** Measures the underlying hardware strain. This is crucial for identifying whether performance bottlenecks are caused by inefficient code or hardware limitations.

Response Time Analysis

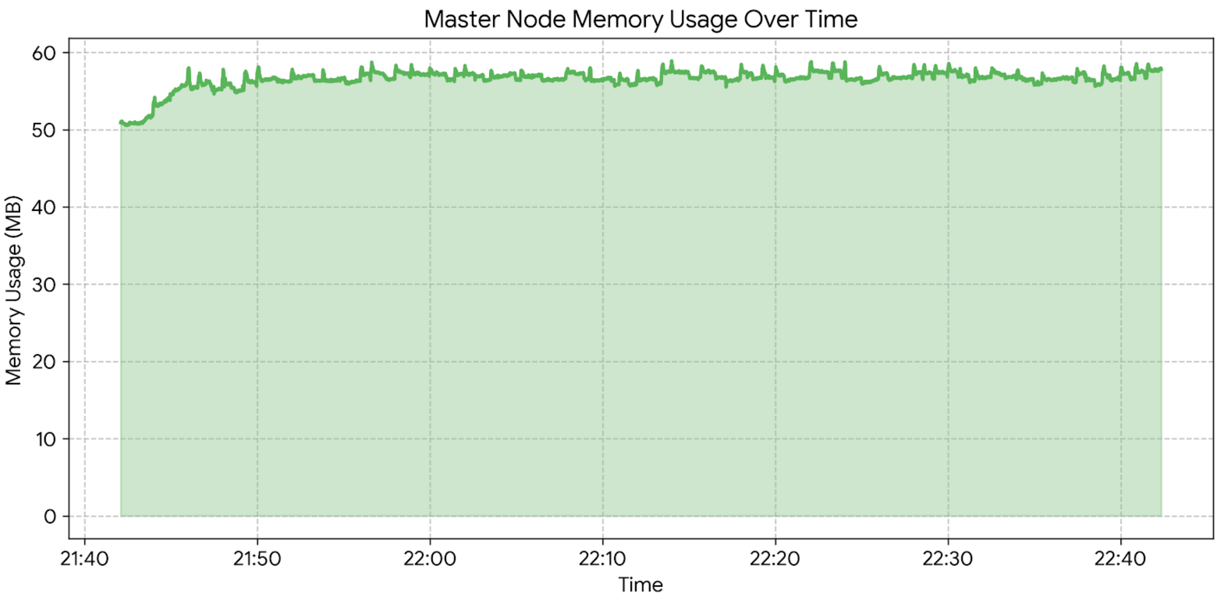
Action	Avg Response Time (ms)	Min Response Time (ms)	Max Response Time (ms)
Initial Login	**1705.80**	**8**	**10531**
Nav to Home Page	**5.14**	**3**	**27**
Download Resource File	**7790.81**	**6**	**19110**
Download Defense App	**18,503.93**	**214**	**33992**

Throughput Analysis

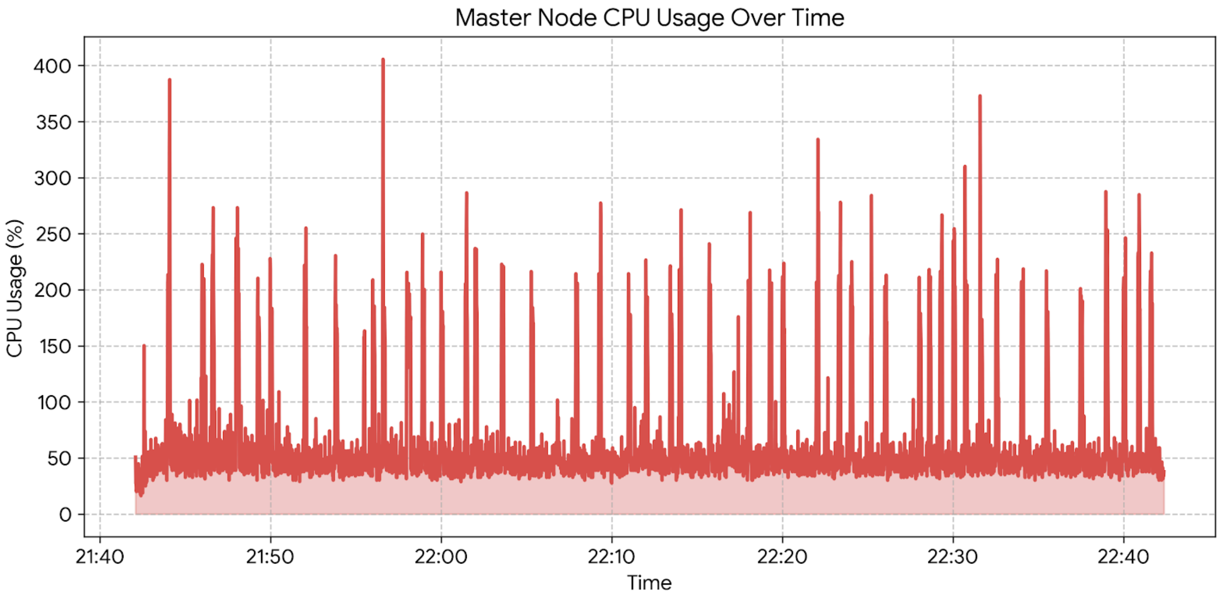
User Load	Avg Throughput (req/sec)	Max Throughput (req/sec)
10	7.05	12
50	25.27	35
100	41.16	57
150	45.67	63
200	36.79	141

System Resource Utilization

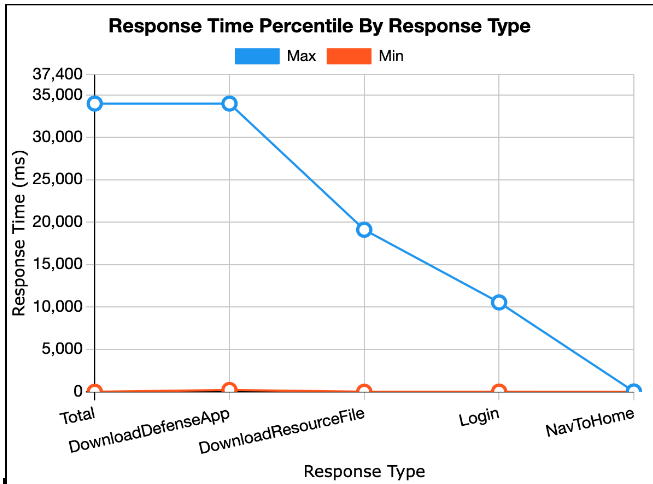
User Load	CPU Usage (%)	Memory Usage (MB)
10	28.28%	50.84
50	41.36%	50.83
100	50.40%	51.58
150	56.77%	53.30
200	62.01%	56.79

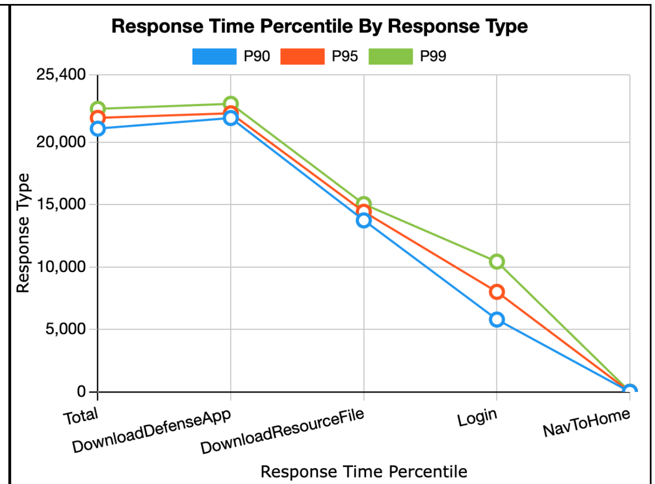


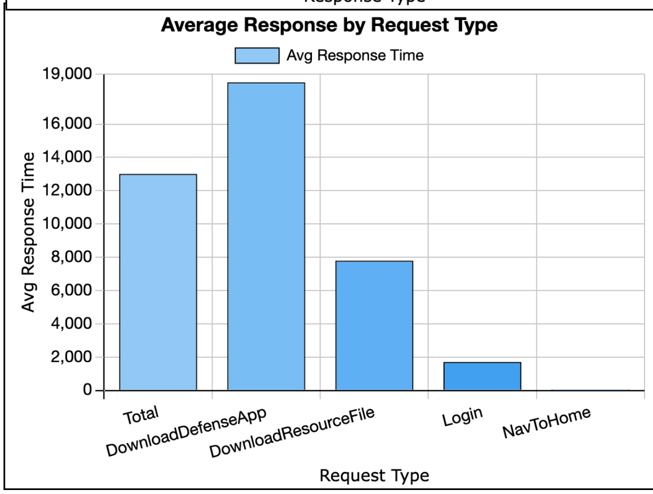
CPU Usage

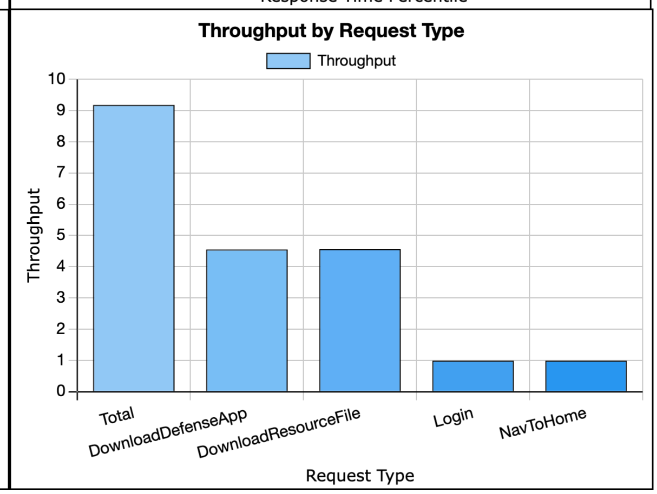


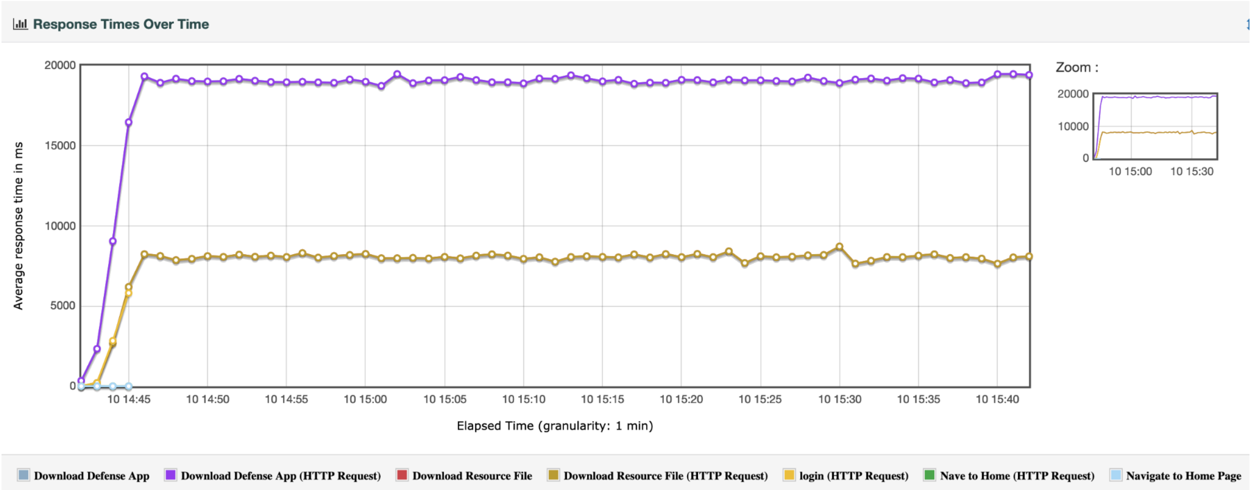
Memory Usage







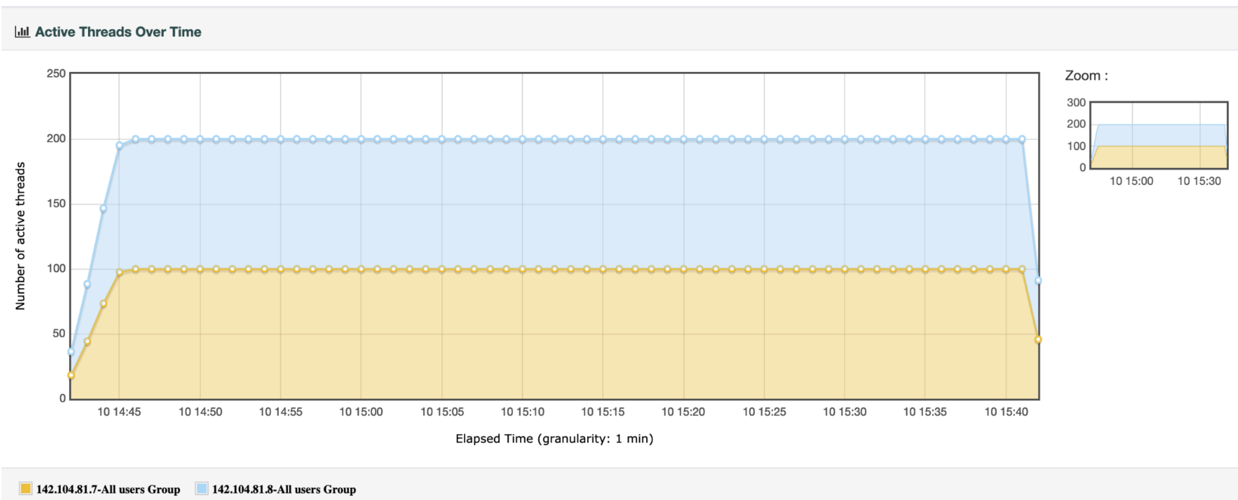




Response Times Over Time

Errors			
Type of error	Number of errors	% in errors	% in all samples

Errors



Active Threads

Active Thread Graph Interpretation

The active threads graph confirmed that the load test ramped up exactly as expected. The test reached 200 concurrent users at about 3.32 minutes, which exactly matched the configured 200 second ramp-up time. The system remained near the full 200-user load for most of the 60-minute test. Near the end, active threads gradually dropped as the test stopped starting new work and existing long-time requests finished. This showcased that the distributed load profile was applied as intended.

CPU Usage Graph Interpretation

The login requests were executed 200 times with an average response time of 1,705.80 ms, The minimum response time was 10,531 ms. There were no errors, and the server handled approximately 0.99 login requests per second across the test duration. The massive gap between the minimum and maximum response times indicated database thread-locking when multiple users attempted to authenticate simultaneously.

Memory Usage Graph Interpretation

The memory usage showed a gradual increase during the test, with it starting out at about 50.98 GB and ending in around 57.80 GB, with a maximum of 58.90 GB. During that stable full-reload period, average memory usage was about 56.80 GB. The graph does not show any sharp crash pattern or uncontrolled memory growth, but it does show that the application used noticeably more memory under sustained load.

Overall Interpretation

The graphs ultimately confirm that the 60-minute test was run correctly and the system remained stable under the 200-user load. Active users were able to reach intended targets, CPU usage reasonably increased with load, and memory usage rose but stayed within a stable range. Since JMeter report showed 0 errors, the main concern is not availability, but performance under heavy download traffic. The resource graphs support the conclusion that file downloads created the biggest sustained load on the system.

System Readiness for Production

Evaluation

The Vega app is very stable. Handling over 33,000 requests without a single crash or 500-level error is a great sign. However, it is not fast enough for production yet. The massive delay on file downloads (18+ seconds) ruins the user experience. Because the server is trying to send large files directly, it eats up CPU and network bandwidth, which slows down the rest of the application.

Recommendations

- **Using a CDN for Large Files:** The 50MB file.jpg and the Defense App ZIP shouldn't be hosted directly on the main server. Moving them to a Content Delivery Network (CDN) or AWS S3 bucket will take the heavy lifting off the Master machine and drastically improve download speeds.
- **Add Database Connection Pooling:** Initial logins averaged 1.7 seconds but spiked up to 10.5 seconds for some users. This means the database is struggling to handle multiple logins at the exact same time. Adding connection pooling (like HikariCP) will fix this bottleneck.

Conclusion

Our load test proved that the Vega Application is incredibly stable, surviving a 200-user stress test with a perfect 100% success rate. CPU usage peaked safely at 62.01%, and memory usage was highly optimized, plateauing at just 56.79 MB under peak load.

However, the application is too slow when handling heavy file downloads. Before rolling this out to the public, the development team needs to offload static file hosting to a CDN and optimize the database connections. Once those two bottlenecks are cleared, the app will be fully ready for production.

Appendix A - JMeter Test Plan

See submitted zip file.

Appendix B - Results File

See .jtl file in submitted zip file.

Appendix A - Screenshots of test configuration and results

1	timeStamp	elapsed	label	responseCoc	responseMes	threadName	dataType	success	failureMessa	bytes	sentBytes	grpThreads	allThreads	URL	Latency	IdleTime	Connect
2	1.7837E+12	3129	142.104.81.9 CPU					TRUE		0	0	0	0	null	0	0	0
3	1.7837E+12	0	142.104.81.9 Network I/O					TRUE		0	0	0	0	null	0	0	0
4	1.7837E+12	4.037E+10	142.104.81.9 Swap					TRUE		0	0	0	0	null	0	0	0
5	1.7837E+12	34564	142.104.81.9 Memory					TRUE		0	0	0	0	null	0	0	0
6	1.7837E+12	20000	142.104.81.9 TCP					TRUE		0	0	0	0	null	0	0	0
7	1.7837E+12	8	142.104.81.9 Disks I/O					TRUE		0	0	0	0	null	0	0	0
8	1.7837E+12	3980	142.104.81.9 CPU					TRUE		0	0	0	0	null	0	0	0
9	1.7837E+12	108708000	142.104.81.9 Network I/O					TRUE		0	0	0	0	null	0	0	0
10	1.7837E+12	4.037E+10	142.104.81.9 Swap					TRUE		0	0	0	0	null	0	0	0
11	1.7837E+12	34568	142.104.81.9 Memory					TRUE		0	0	0	0	null	0	0	0
12	1.7837E+12	20000	142.104.81.9 TCP					TRUE		0	0	0	0	null	0	0	0
13	1.7837E+12	0	142.104.81.9 Disks I/O					TRUE		0	0	0	0	null	0	0	0
14	1.7837E+12	5875	142.104.81.9 CPU					TRUE		0	0	0	0	null	0	0	0
15	1.7837E+12	5576706000	142.104.81.9 Network I/O					TRUE		0	0	0	0	null	0	0	0
16	1.7837E+12	4.037E+10	142.104.81.9 Swap					TRUE		0	0	0	0	null	0	0	0
17	1.7837E+12	34573	142.104.81.9 Memory					TRUE		0	0	0	0	null	0	0	0
18	1.7837E+12	20000	142.104.81.9 TCP					TRUE		0	0	0	0	null	0	0	0
19	1.7837E+12	8	142.104.81.9 Disks I/O					TRUE		0	0	0	0	null	0	0	0
20	1.7837E+12	3764	142.104.81.9 CPU					TRUE		0	0	0	0	null	0	0	0
21	1.7837E+12	2811770000	142.104.81.9 Network I/O					TRUE		0	0	0	0	null	0	0	0
22	1.7837E+12	4.037E+10	142.104.81.9 Swap					TRUE		0	0	0	0	null	0	0	0
23	1.7837E+12	34577	142.104.81.9 Memory					TRUE		0	0	0	0	null	0	0	0
24	1.7837E+12	20000	142.104.81.9 TCP					TRUE		0	0	0	0	null	0	0	0
25	1.7837E+12	16	142.104.81.9 Disks I/O					TRUE		0	0	0	0	null	0	0	0
26	1.7837E+12	1246	142.104.81.9 CPU					TRUE		0	0	0	0	null	0	0	0
27	1.7837E+12	6242000	142.104.81.9 Network I/O					TRUE		0	0	0	0	null	0	0	0
28	1.7837E+12	4.037E+10	142.104.81.9 Swap					TRUE		0	0	0	0	null	0	0	0
29	1.7837E+12	34576	142.104.81.9 Memory					TRUE		0	0	0	0	null	0	0	0
30	1.7837E+12	21000	142.104.81.9 TCP					TRUE		0	0	0	0	null	0	0	0
31	1.7837E+12	0	142.104.81.9 Disks I/O					TRUE		0	0	0	0	null	0	0	0
32	1.7837E+12	3370	142.104.81.9 CPU					TRUE		0	0	0	0	null	0	0	0
33	1.7837E+12	111705000	142.104.81.9 Network I/O					TRUE		0	0	0	0	null	0	0	0
34	1.7837E+12	4.037E+10	142.104.81.9 Swap					TRUE		0	0	0	0	null	0	0	0
35	1.7837E+12	34592	142.104.81.9 Memory					TRUE		0	0	0	0	null	0	0	0

MASTER_IP = 142.104.81.9

WORKER1_IP = 142.104.81.7

WORKER2_IP = 142.104.81.8

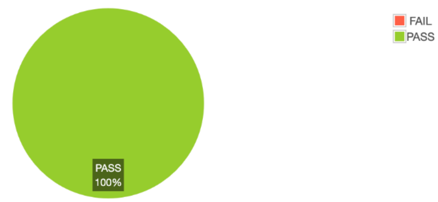
Test and Report information

Source file	"results.jtl"
Start Time	"7/10/26, 2:42 PM"
End Time	"7/10/26, 3:42 PM"
Filter for display	""

APDEX (Application Performance Index)

Apdex	T (Toleration threshold)	F (Frustration threshold)	Label
0.037	**500 ms**	**1 sec 500 ms**	**Total**
0.011	500 ms	1 sec 500 ms	Download Defense App (HTTP Request)
0.011	500 ms	1 sec 500 ms	Download Defense App
0.047	500 ms	1 sec 500 ms	Download Resource File
0.047	500 ms	1 sec 500 ms	Download Resource File (HTTP Request)
0.610	500 ms	1 sec 500 ms	login (HTTP Request)
1.000	500 ms	1 sec 500 ms	Navigate to Home Page
1.000	500 ms	1 sec 500 ms	Nave to Home (HTTP Request)

Requests Summary



Statistics

Requests	Executions			Response Times (ms)							Throughput	Network (KB/sec)	
Label	#Samples	FAIL	Error %	Average	Min	Max	Median	90th pct	95th pct	99th pct	Transactions/s	Received	Sent
Total	**33220**	**0**	**0.00%**	**13008.41**	**3**	**33992**	**15173.50**	**21095.00**	**21959.00**	**22685.99**	**9.18**	**91807.44**	**3.75**
Download Defense App	16437	0	0.00%	18503.93	214	33992	18825.00	21940.40	22323.10	23086.00	4.55	91722.22	1.97
Download Defense App (HTTP Request)	16437	0	0.00%	18503.93	214	33992	18825.00	21940.40	22323.10	23086.00	4.55	91722.25	1.97
Download Resource File	16382	0	0.00%	7790.81	6	19110	7500.00	13745.10	14430.70	15047.34	4.56	233.48	1.77
Download Resource File (HTTP Request)	16382	0	0.00%	7790.81	6	19110	7500.00	13745.10	14430.70	15047.34	4.56	233.48	1.77
login (HTTP Request)	200	0	0.00%	1705.80	8	10531	207.00	5799.50	8016.05	10437.54	0.99	0.72	0.24
Nave to Home (HTTP Request)	200	0	0.00%	5.14	3	27	5.00	6.00	7.95	20.00	0.99	2.01	0.34
Navigate to Home Page	200	0	0.00%	5.14	3	27	5.00	6.00	7.95	20.00	0.99	2.01	0.34